Worksheet 5

Time series

Complete exercises

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1. Core applications

Master the essential definitions and calculations from the course.

Core application 10 min

Exercise 1 – Components and models

1. Match long-run trend, recession, recurring Q4 peak and exceptional shock to T_t, C_t, S_t or R_t.
2. Write the full additive and multiplicative models.
3. Explain when multiplicative seasonality is more plausible than additive seasonality.

Core application 20 min

Exercise 2 – Moving averages and centring

Quarterly demand over two years is

$$80,\ 100,\ 120,\ 140,\ 88,\ 110,\ 132,\ 154.$$

Describe the main pattern, calculate the five 4-quarter moving averages, centre them for periods 3 to 6, and explain why an even-order moving average must be centred.

2. Assessment and exam practice

Practise independently with assessment and final-exam style questions.

Assessment / exam practice 25 min

Exercise 3 – Multiplicative decomposition and forecasting

Year	Q1	Q2	Q3	Q4
2005	82	98	120	150
2006	91	111	137	171
2007	104	126	154	194

Given $\widehat{T_t} = 6.343t + 86.939$ and seasonal coefficients $0.789, 0.903, 1.052, 1.255$, interpret the trend slope, check normalization, forecast the four quarters of 2008, and explain multiplicative deseasonalization.

Assessment / exam practice 65 min

Exercise 5 — Full seasonal decomposition — exam style

A firm observes quarterly sales over three years. Use the multiplicative model $Y_t = T_t \times S_t$, without explicit cycle or irregular component. Do not round intermediate values.

Année / Year	Q1	Q2	Q3	Q4
2019	40	52	48	68
2020	44	58	53	76
2021	50	65	60	84

1. Plot the series and calculate all nine 4-quarter moving averages.
2. Centre the moving averages (t=3 to 10). Build the t, CMA4, t^2, t×CMA4 table.
3. Estimate trend $T_t = at + b$ by OLS on the eight centred moving averages.
4. Calculate Y_t/T_t for all 12 quarters, the four seasonal means and normalised factors summing to 4.
5. Forecast all four quarters of 2022 (t=13 to 16), first deseasonalised, then seasonalised.

3. Optional advanced practice

Explore additional interpretations and professional applications.

Advanced practice 15 min

Exercise 4 – Forecast evaluation

For this exercise, suppose actual 2008 values were $136, 162, 188, 245$. Using the Exercise 3 forecasts, calculate forecast errors, MAE and RMSE, and explain the value of a separate validation period and a naive benchmark.

Advanced practice 20 min

Extra 1 — Forecast validation and structural break

Actual monthly demand is 100, 104, 108, 150, 154 and 158. A trend model fitted on the first three months predicts 100, 104, 108, 112, 116 and 120.

1. Calculate forecast errors for months 4–6.

2. Calculate MAE and mean error over the validation period.
3. Diagnose the model.
4. Propose a responsible forecasting response.

Use error = actual − forecast. A persistently positive error signals systematic underforecasting.